

INFO8006 Introduction to Artificial Intelligence

Exercise session 4: Probabilistic reasoning

RECAP

Bayesian networks

With a partially observable environment, an agent reasons about the state s from its percepts e with the joint distribution $P(s, e) = P(s)P(e | s)$. When s and e are assignments of many random variables $X_1, \dots, X_n$, a *Bayesian network* represents this distribution compactly. It is a directed acyclic graph with one node per variable, each node X_i annotated with a conditional distribution $\mathbf{P}(X_i | \text{parents}(X_i))$, given as a *conditional probability table* (CPT) for discrete variables. The topology asserts the *local Markov property*, $X_i \perp \text{non-descendants}(X_i) | \text{parents}(X_i)$, hence the factorization

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i)).$$

Naive Bayes is the network in which a cause C is the only parent of each effect E_i .

d-separation

$X_i \perp X_j | Z_1, \dots, Z_m$ is guaranteed if every undirected path between X_i and X_j is *inactive*; if one path is *active*, independence is not guaranteed. A path is active iff each of its triples is active:

Triple		Active iff
Cascade	$A \rightarrow B \rightarrow C$ or $A \leftarrow B \leftarrow C$	B unobserved
Common parent	$A \leftarrow B \rightarrow C$	B unobserved
v-structure	$A \rightarrow B \leftarrow C$	B or a descendant of B observed

Given its *Markov blanket*, *i.e.* its parents, its children and the other parents of its children, a variable is independent of all the others.

Inference

Given *evidence* variables $\mathbf{E}$ observed as $\mathbf{e}$, a *query* variable Q and *hidden* variables $\mathbf{H}$, inference computes

$$\mathbf{P}(Q | \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \quad Z = \sum_q \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}),$$

with the joint entries written as products of CPT entries. *Inference by enumeration* evaluates the sums depth-first, in $O(n)$ space and $O(d^n)$ time for n variables with d values. *Variable elimination* evaluates them right to left and stores the intermediate *factors*, multi-dimensional arrays initialized with the CPTs given the evidence. It alternates the *join* of factors, their pointwise product over the union of their variables, and the *elimination* of a hidden variable,

summed out of the join of the factors that mention it. Its cost is that of the largest factor, which depends on the elimination ordering; finding the best ordering is NP-hard. A hidden variable H is *irrelevant*, and can be removed, unless $H \in \text{ancestors}(\{Q\} \cup \mathbf{E})$.

Parameter learning

The CPTs can be chosen in a *parameterized* family $\mathbf{P}(X \mid \theta)$, whose parameters are estimated from N i.i.d. observations $\mathbf{d} = \{x_1, \dots, x_N\}$. The *maximum likelihood estimate* (MLE) maximizes the *likelihood* $P(\mathbf{d} \mid \theta) = \prod_{j=1}^N P(x_j \mid \theta)$, or the *log-likelihood* $L(\theta) = \log P(\mathbf{d} \mid \theta)$, often by setting $\frac{\partial L}{\partial \theta} = 0$:

$$\theta_{\text{MLE}}^* = \arg \max_{\theta} P(\mathbf{d} \mid \theta).$$

Bayesian parameter learning treats θ as a random variable with a *prior* $\mathbf{P}(\theta)$, updated into the *posterior* $\mathbf{P}(\theta \mid \mathbf{d}) \propto \mathbf{P}(\mathbf{d} \mid \theta)\mathbf{P}(\theta)$. A new observation x is then predicted by the *posterior predictive* $p(x \mid \mathbf{d}) = \int p(x \mid \theta)p(\theta \mid \mathbf{d}) d\theta$, and by the *prior predictive* $p(x) = \int p(x \mid \theta)p(\theta) d\theta$ before any data. When the posterior is intractable, the *maximum a posteriori* (MAP) estimate replaces it by its mode,

$$\theta_{\text{MAP}}^* = \arg \max_{\theta} P(\theta \mid \mathbf{d}) = \arg \max_{\theta} P(\mathbf{d} \mid \theta)P(\theta).$$

Cheat sheet for Gaussian distributions

The Gaussian density of mean μ and variance σ^2 is

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{a} \\ \mathbf{b} \end{pmatrix}, \begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^T & \mathbf{B} \end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}) \\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}) \\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{CB}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{CB}^{-1}\mathbf{C}^T) \\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

Conversely, if $\mathbf{x}$ and $\mathbf{y}$ have the Gaussian distributions

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{m}, \mathbf{P}) \\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R}), \end{aligned}$$

then their joint distribution is Gaussian with

$$p\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}\right) = \mathcal{N}\left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \middle| \begin{pmatrix} \mathbf{m} \\ \mathbf{H}\mathbf{m} + \mathbf{u} \end{pmatrix}, \begin{pmatrix} \mathbf{P} & \mathbf{P}\mathbf{H}^T \\ \mathbf{H}\mathbf{P} & \mathbf{H}\mathbf{P}\mathbf{H}^T + \mathbf{R} \end{pmatrix}\right).$$

In session: exercises 1, 2.

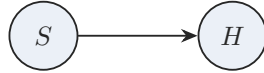
Exercise 1 Nuclear power plant (AIMA ch. 13)

A monitoring program supervises the core of a nuclear power station. An alarm sounds when a temperature gauge, which measures the temperature of the core, exceeds a given threshold, and the program reasons about the actual temperature from the gauge and the alarm, either of which may be faulty. Consider the Boolean variables A (alarm sounds), F_A (alarm is faulty), and F_G (gauge is faulty) and the multi-valued nodes G (gauge reading) and T (actual core temperature).

- (a) Draw a Bayesian network for this domain, given that the gauge is more likely to fail when the core temperature gets too high.
- (b) From your network topology, can you conclude $T \perp F_A \mid F_G$ and $T \perp F_A \mid F_G, A$?
- (c) Suppose there are just two possible actual and measured temperatures: low (l) and high (h). The probability that the gauge gives the correct temperature is x when it is working, but y when it is faulty. Give the conditional probability table associated with G .
- (d) Suppose the alarm is always triggered by high measured temperatures, unless it is faulty, in which case it never sounds. Give the conditional probability table associated with A .
- (e) Suppose the gauge is not faulty and the alarm is triggered. Calculate an expression for the probability that the temperature of the core is too high, in terms of the various conditional probabilities in the network.

Exercise 2 Energy of an inference server

A scheduling program manages a data center in which a GPU server answers the requests sent to a language model. To plan the power supply, the scheduler models the energy the server consumes as a function of its load, with the following network



where S is a random variable denoting the number of requests the server receives in an hour, and H a random variable denoting the energy, in kilowatt-hours, that it consumes during that hour. The scheduler chooses the following relations for each variable

$$P(S = s) = \text{Poisson}(S = s; \lambda) = \frac{\lambda^s e^{-\lambda}}{s!} \quad (1)$$

$$H = \omega S + H_0 + \mathcal{N}(0, \sigma^2) \quad (2)$$

where H_0 is the energy the server consumes when idle, which is known, and ω is the energy consumed per request.

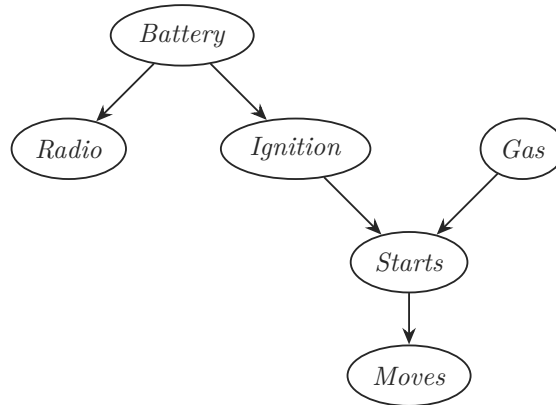
- (a) Identify the parameters of the Bayesian network and give the joint log-likelihood of one pair (s_i, h_i) given the model.
- (b) The scheduler collects a set $\mathbf{d} = \{(s_i, h_i)\}_{i=1}^N$ of independent observations. Derive the maximum likelihood estimate of the parameters.
- (c) The scheduler has just started to collect observations, and maximum likelihood can overfit when they are few. It therefore treats the parameter λ , about which it has no strong knowledge, as uncertain. Using a prior $\lambda \sim \text{Gamma}(\alpha, \beta)$, what would be the posterior distribution $p(\lambda \mid \mathbf{s})$, with $\mathbf{s} = \{s_i\}_{i=1}^N$?

Hint: $\text{Gamma}(x \mid \alpha, \beta) = \frac{1}{Z} x^{\alpha-1} e^{-\beta x}$, with Z a normalizing constant.

- (d) Knowing that the mean, the variance and the mode of $\text{Gamma}(\alpha, \beta)$ are respectively $\frac{\alpha}{\beta}$, $\frac{\alpha}{\beta^2}$ and $\frac{\alpha - 1}{\beta}$, interpret how they evolve w.r.t. N between the prior and the posterior.

Exercise 3 Car diagnosis (AIMA ch. 13)

Consider the following Bayesian network, which describes some features of a car's electrical system and engine. Each variable is Boolean, and the true value indicates that the corresponding aspect of the vehicle is in working order.



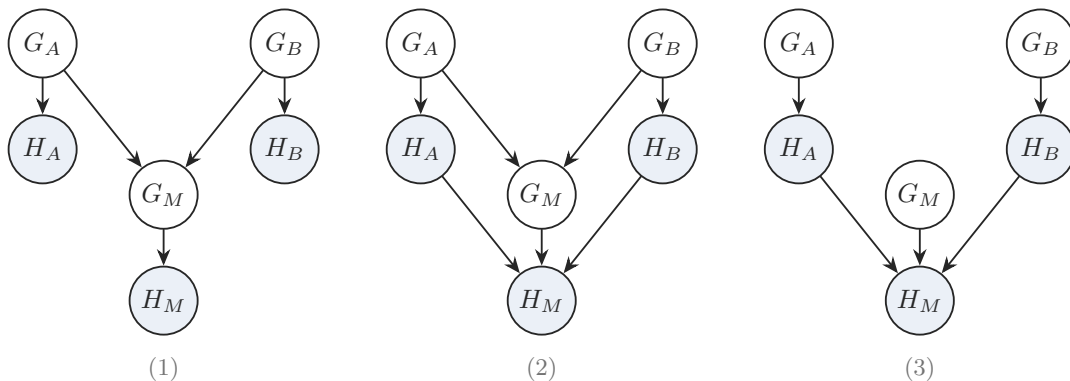
- Extend the network with the Boolean variables IcyWeather and StarterMotor.
- According to your knowledge of cars, give reasonable conditional probability tables for all the nodes.
- How many independent values are contained in the joint probability distribution for eight Boolean nodes, assuming that no conditional independence relations are known to hold among them?
- How many independent probability values do your network tables contain?

Exercise 4 Evolutionary model merging (AIMA ch. 13)

Evolutionary model merging builds a new language model M from two parent models A and B : each component of M is copied from one of the parents. Consider one component, and let G_x be the language it was mostly trained on in model x , with possible values e (English) or f (French). A common *hypothesis* is that the default language of a model follows from this component by a simple mechanism: the language H_x in which model x answers by default is the same as G_x with some probability s .

Furthermore, M copies the component from either parent with equal probability, and a random perturbation flips it with a small nonzero probability m .

- (a) Which of the following networks claim that $\mathbf{P}(G_M \mid G_A, G_B) = \mathbf{P}(G_M)$? Shaded variables are observed.



- (b) Which of the networks make independence claims that are consistent with the hypothesis about the default language?
- (c) Write down the CPT for the G_M node in the first network, in terms of s and m .
- (d) Suppose that $P(G_A = f) = P(G_B = f) = q$. In the first network, derive an expression for $P(G_M = f)$ in terms of s , m and q .
- (e) Over many generations of merges, we expect the distribution of the component to be the same in the parents and in the merged models. Use this to calculate the value of q . Only about 10 % of the merged models answer in French by default: explain why the hypothesis described at the beginning of this question must be wrong.

Exercise 5 Gaussian parameter learning with known σ^2

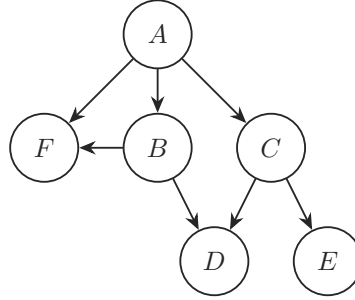
You have in your possession a high-tech laser that emits light of intensity μ_l with great precision. You want to use it to test a new light-intensity measurement device. You assume that your measurements y_i follow a Gaussian distribution $\mathcal{N}(\mu_y, \sigma^2)$ with $\mu_y \sim \mathcal{N}(\mu_l, \sigma_l^2)$ corresponding to the model of your laser given in the manual. You assume that σ^2 is fixed and that you can determine it.

You have access to a calibration set $\mathcal{Y}_c$ of C observations, and you want to predict a new measurement y of your device. You want to study how much $\mathcal{Y}_c$ reduces your uncertainty about this measurement.

- (a) Compute the prior and posterior predictive distributions $P(y)$ and $P(y | \mathcal{Y}_c)$ if you have access to one observation, *i.e.* $\mathcal{Y}_c = \{y_o\}$.
- (b) Looking at your predictive uncertainty, what is the benefit of incorporating $\mathcal{Y}_c$?

Exercise 6 Independence and variable elimination

Considering the Bayesian network below, which of the following statements are enforced by the network structure?

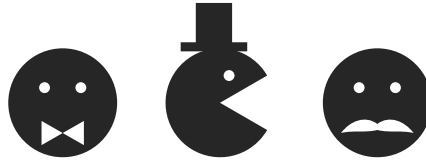


- (a) $\mathbf{P}(B, C) = \mathbf{P}(B)\mathbf{P}(C)$
- (b) $\mathbf{P}(B, C | A) = \mathbf{P}(B | A)\mathbf{P}(C | A)$
- (c) $\mathbf{P}(F, E) = \mathbf{P}(F)\mathbf{P}(E)$
- (d) $\mathbf{P}(F | A, D, E) = \mathbf{P}(F | A, D)$
- (e) $\mathbf{P}(B, E) = \sum_{a,c,d,f} P(a)\mathbf{P}(B | a)P(c | a)\mathbf{P}(d | B, c)\mathbf{P}(E | c)\mathbf{P}(f | a, B)$

For the same network, use inference by variable elimination to compute $\mathbf{P}(E | A = 1, B = 1)$.

Exercise 7 Recognizing Pacman (CS188, Spring 2014)

The ghosts of the maze are controlled by a program that watches the maze through cameras and must recognize Pacman in their pictures. It learns to do so from a set of pictures of Pacman, and of ghosts as negative examples.



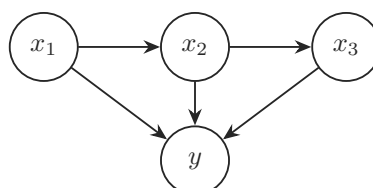
Because the pictures are black and white, and were taken from various angles, the program identifies Pacman from salient features: the presence of a bowtie B , hat H or mustache M . The following table summarizes the content of the pictures. Each feature takes realization in $\{0, 1\}$, where 0 and 1 mean the feature is respectively absent and present. The subject of the picture is described by a random variable $S \in \{0, 1\}$, where 0 is a ghost and 1 is Pacman.

B	H	M	S
0	0	0	1
1	0	0	0
1	0	1	1
1	1	0	0
0	1	0	0
1	1	1	1

- Suppose the program is a naive Bayes classifier. Draw the Bayesian network that represents the dependencies between S , B , H and M for the program.
- Write the Bayesian classification rule for this problem, *i.e.* the formula that given a data point (b, h, m) returns the most likely subject. Write the formula in terms of conditional and prior probabilities. What does the formula become under the assumptions of the program?
- What are the parameters of this model? Give estimates of these parameters according to the pictures of the table.
- Pacman eventually shows up in front of a camera wearing a bowtie, but no hat or mustache. Will the program recognize him?

Exercise 8 Predict the leaderboard score

A leaderboard ranks language models by an overall score. The Bayesian network below represents how this score is computed. In this model, x_1 , x_2 and x_3 respectively denote the scores obtained by a model on a validation set, on a public benchmark and on a held-out test set. The three evaluations share the same harness, and a model that does well on one tends to do well on the next. In particular, $x_2 \sim \mathcal{N}(a_1x_1 + \mu_2, \sigma_2^2)$ and $x_3 \sim \mathcal{N}(a_2x_2 + \mu_3, \sigma_3^2)$. Finally, $y \sim \mathcal{N}(a_3x_1 + a_4x_2 + a_5x_3 + \mu_y, \sigma_y^2)$ stands for the overall score, which is a linear combination of the three scores plus some Gaussian noise due to rounding errors. Answer the following questions about this model.



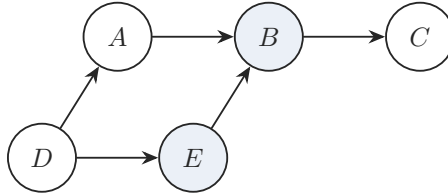
- (a) Assuming the parameters of the model are known, what is the expected value of y given x_1 and x_2 ?
- (b) Suppose now that the parameters are unknown, and that the scores x_3 on the held-out test set are not observed. Given a learning set $d = \{(x_{i,1}, x_{i,2}, y_i)\}$ of N independent and identically distributed points, derive the maximum likelihood estimate of the parameters of $p(y \mid x_1, x_2)$.

Exercise 9 Heteroscedastic linear regression

In linear regression with Gaussian noise of constant variance, $y_i \sim \mathcal{N}(w^T x_i, \sigma^2)$, the maximum likelihood estimate of the weights is $w_{\text{MLE}} = (X^T X)^{-1} X^T Y$, where the rows of X are the inputs x_i^T and $Y = (y_i)$. What does this estimate become if the noise is different for each sample, *i.e.* $y_i \sim \mathcal{N}(w^T x_i, \sigma_i^2)$ with known variances σ_i^2 ?

QUIZ

- The Normal distribution $\mathcal{N}(\mu, \sigma)$ is described by the density function
 - ☐ $p(x) = \frac{1}{\sigma} \exp(-(z + \exp(-z)))$, with $z = \frac{x-\mu}{\sigma}$.
 - ☐ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
 - ☐ $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
 - ☐ $p(x) = 1 / \pi\sigma \left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]$.
- Consider the Bayesian network shown below. We want to infer $\mathbf{P}(A \mid b, e)$ where A is the query variable, B and E are evidence variables, and C and D are hidden variables. Which of the following statements is true?



- ☐ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b) \mathbf{P}(b \mid A) P(b \mid e) \mathbf{P}(A \mid d) P(e \mid d) P(d)$
 - ☐ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b) \mathbf{P}(b \mid A, e) \mathbf{P}(A \mid d) P(e \mid d)$
 - ☐ $\mathbf{P}(A \mid b, e) \propto \sum_c \sum_d P(c \mid b) \mathbf{P}(b \mid A, e) \mathbf{P}(A \mid d) P(e \mid d) P(d)$
 - ☐ $\mathbf{P}(A \mid b, e) \propto \sum_b \sum_e P(c \mid b) \mathbf{P}(A \mid b, e) \mathbf{P}(A \mid d) P(e \mid d) P(d)$
- In a Bayesian network
 - ☐ independence between two variables is guaranteed if all paths connecting them are inactive.
 - ☐ dependence between two variables can be guaranteed using d-separation.
 - ☐ a path between two nodes is active if it contains at least one active triple.
 - ☐ a cascade triple is active if the center node is observed.
 - Consider a Bayesian network with a single binary variable, following a Bernoulli distribution. If we observe T positive realizations and F negative ones,
 - ☐ the maximum likelihood estimate of the positive probability is $\frac{T}{F}$.
 - ☐ the maximum likelihood estimate of the positive probability is $\frac{T+F}{T}$.
 - ☐ the maximum likelihood estimate of the positive probability is $\frac{T}{T+F}$.
 - ☐ None of the above.